

1.Singleton Pattern

```
using System;

class Logger
{
    private static Logger instance;

    private Logger()
    {
        Console.WriteLine("Logger instance created");
    }

    public static Logger GetInstance()
    {
        if (instance == null)
        {
            instance = new Logger();
        }
        return instance;
    }

    public void Log(string message)
    {
        Console.WriteLine("Log: " + message);
    }
}

class SingletonPatternExample
{
    static void Main(string[] args)
    {
        Logger logger1 = Logger.GetInstance();
        Logger logger2 = Logger.GetInstance();

        logger1.Log("First message");
        logger2.Log("Second message");

        Console.WriteLine(logger1 == logger2);
    }
}
```

```
}  
}
```

Output

Logger instance created
Log: First message
Log: Second message
True

2.Factory Method

```
using System;
```

```
interface IDocument  
{  
    void Open();  
}
```

```
class WordDocument : IDocument  
{  
    public void Open()  
    {  
        Console.WriteLine("Opening Word Document");  
    }  
}
```

```
class PdfDocument : IDocument  
{  
    public void Open()  
    {  
        Console.WriteLine("Opening PDF Document");  
    }  
}
```

```
class ExcelDocument : IDocument  
{  
    public void Open()  
    {  
        Console.WriteLine("Opening Excel Document");  
    }  
}
```

```
    }  
}
```

```
abstract class DocumentFactory  
{  
    public abstract IDocument CreateDocument();  
}
```

```
class WordFactory : DocumentFactory  
{  
    public override IDocument CreateDocument()  
    {  
        return new WordDocument();  
    }  
}
```

```
class PdfFactory : DocumentFactory  
{  
    public override IDocument CreateDocument()  
    {  
        return new PdfDocument();  
    }  
}
```

```
class ExcelFactory : DocumentFactory  
{  
    public override IDocument CreateDocument()  
    {  
        return new ExcelDocument();  
    }  
}
```

```
class FactoryMethodPatternExample  
{  
    static void Main(string[] args)  
    {  
        DocumentFactory wordFactory = new WordFactory();  
        IDocument word = wordFactory.CreateDocument();  
        word.Open();  
  
        DocumentFactory pdfFactory = new PdfFactory();
```

```
        IDocument pdf = pdfFactory.CreateDocument();  
        pdf.Open();  
  
        DocumentFactory excelFactory = new ExcelFactory();  
        IDocument excel = excelFactory.CreateDocument();  
        excel.Open();  
    }  
}
```

Output

Opening Word Document
Opening PDF Document
Opening Excel Document